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Weight-Optimized Ensemble Deep Learning for Accurate Mammographic Image Classification

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1. Introduction

Challenges

- Breast cancer is one of the most common cancers affecting women worldwide.
- Early detection and accurate staging of breast cancer can significantly improve survival rates and quality of life for patients.

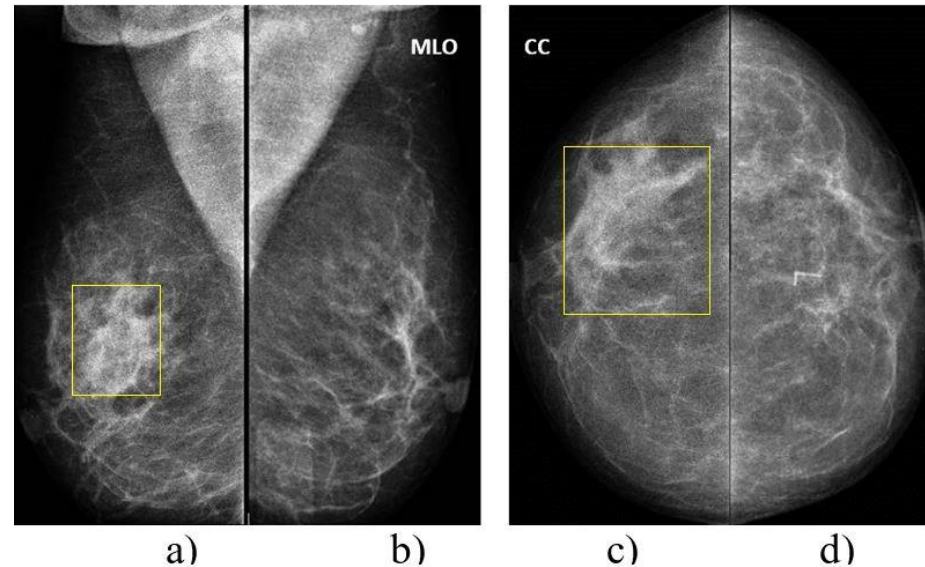


Fig 1. Images on 2 views a) RMLO b) LMLO c) RCC d) LCC with the mass location localized in images a) and c)

1. Introduction

Challenges

Breast cancer (BC) predicted has affected 2.3 million women worldwide in 2020, with 685,000 fatalities. Breast cancer had been detected in 7.8 million females in the previous five years as of 2020, making it the most common cancer on the planet. It causes more disability adjusted life years (DALYs) in women worldwide than any other form of cancer[1].

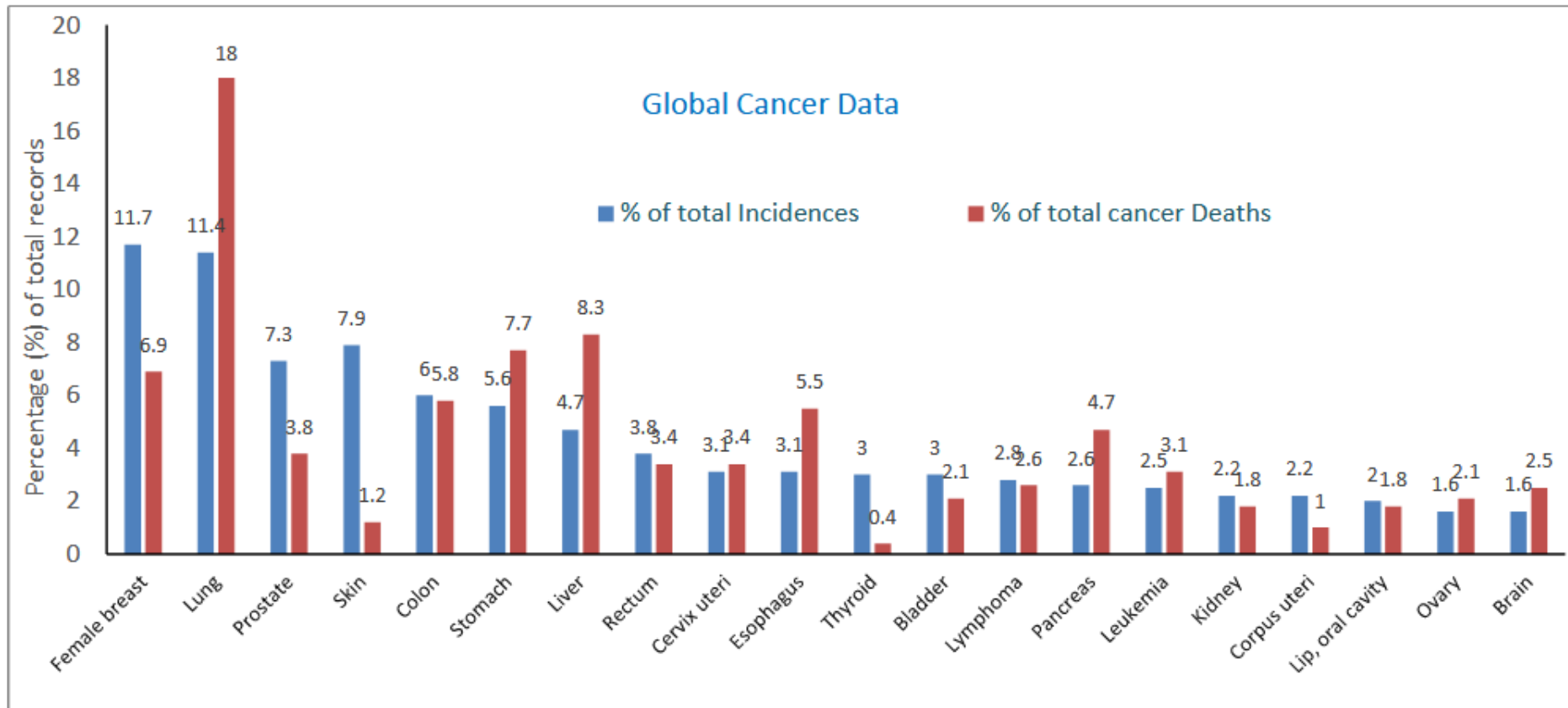
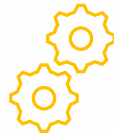


Fig 2. Cancer statistics in the world (Source: Globocan 2023)

1. Introduction

Targets:

Install and experiment with the model, test, evaluate and compare the results.

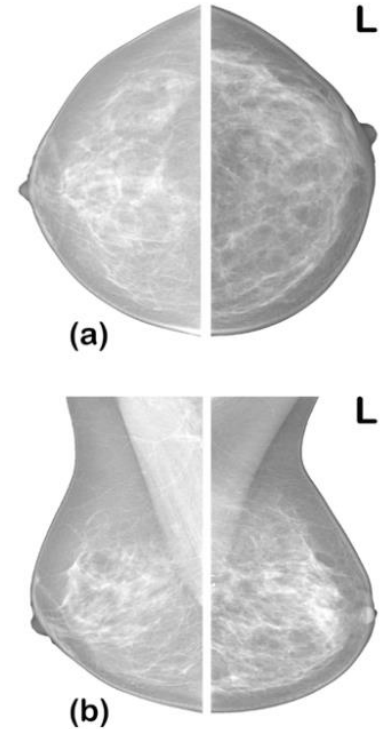
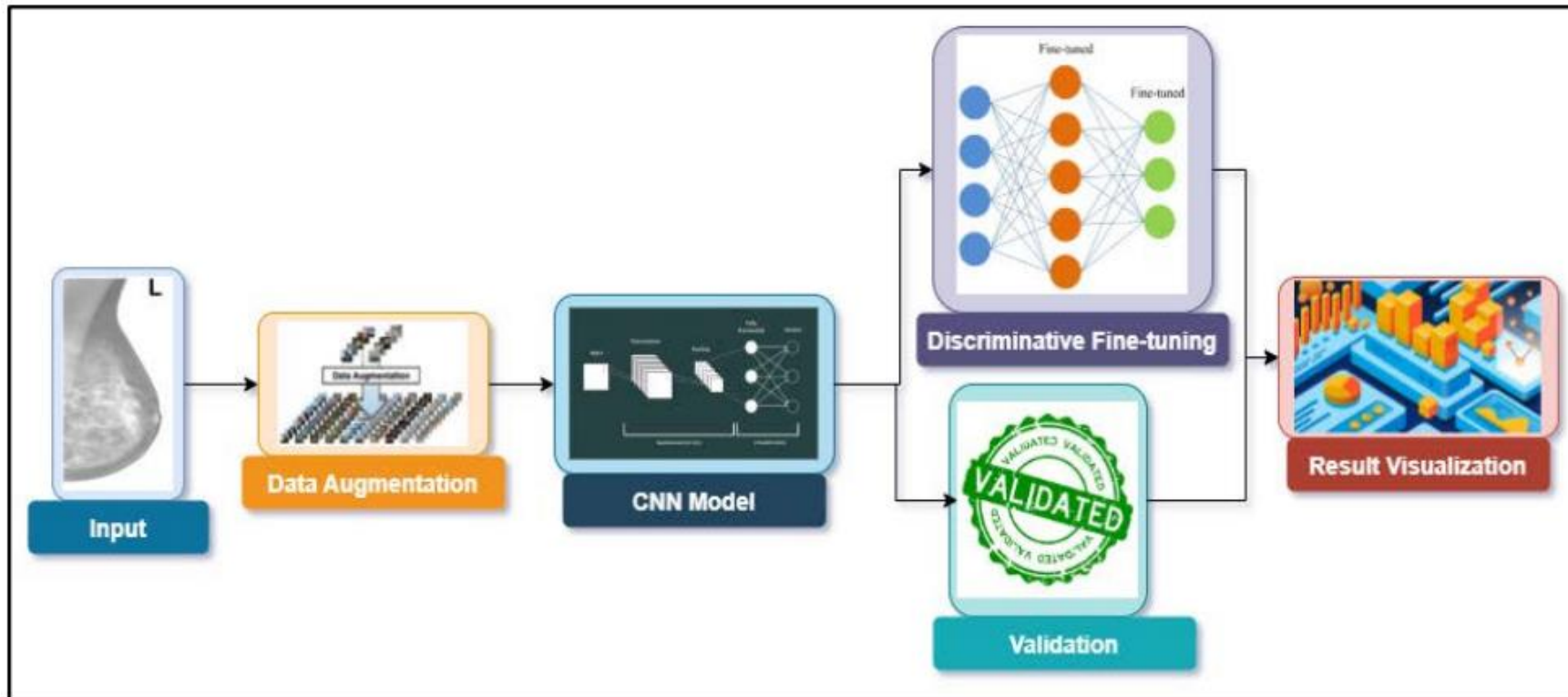


Detection and classification on mammograms based on deep learning techniques



Optimizing weights of ensemble model to reduce the classification error rate

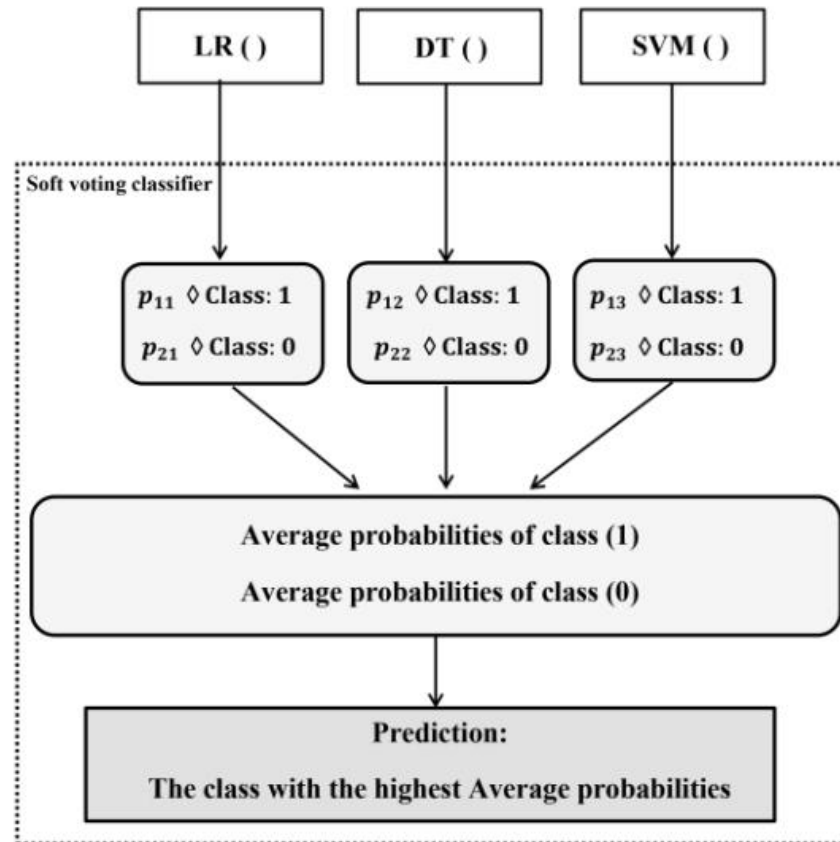
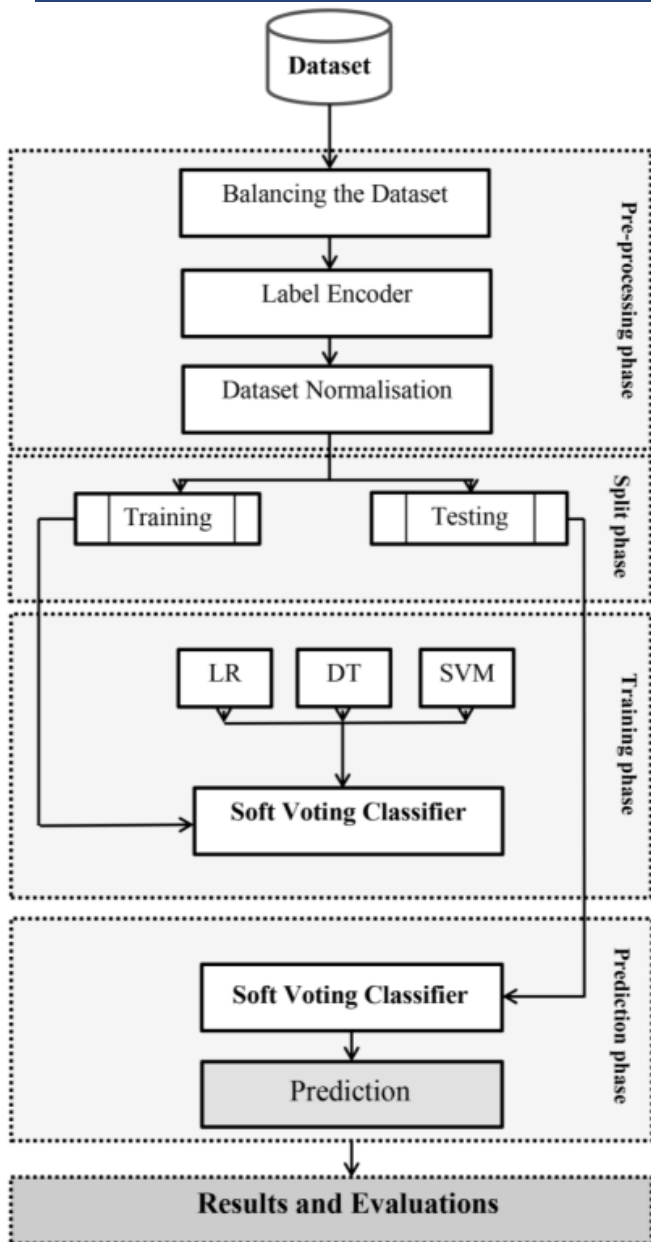
2. Related works



- Architecture network: AlexNet, VGG, SqueezeNet, ResNet, DenseNet
- Dataset: CBIS-DDSM
- Accuracy: 99.8%

Fig 3. Performance Evaluation of Deep Learning Models on Mammogram Classification Using Small Dataset [3]

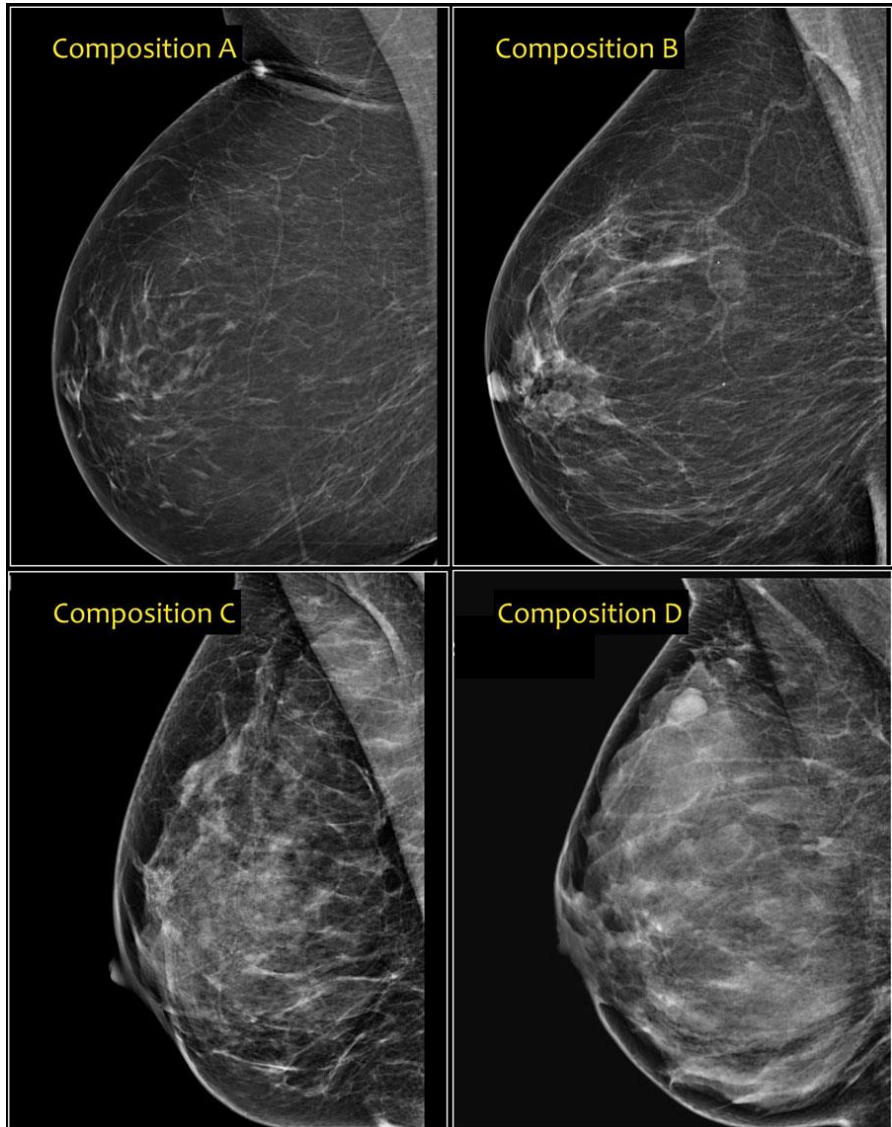
2. Related works



- Ensemble method: Decision Tree, Logistic Regression and Support Vector Machine
- Dataset: WDBC
- Accuracy: 98.25% (Hard voting)

Fig 4. Breast Cancer Prediction Using Soft Voting Classifier Based on Machine Learning Models [4]

2. Related works

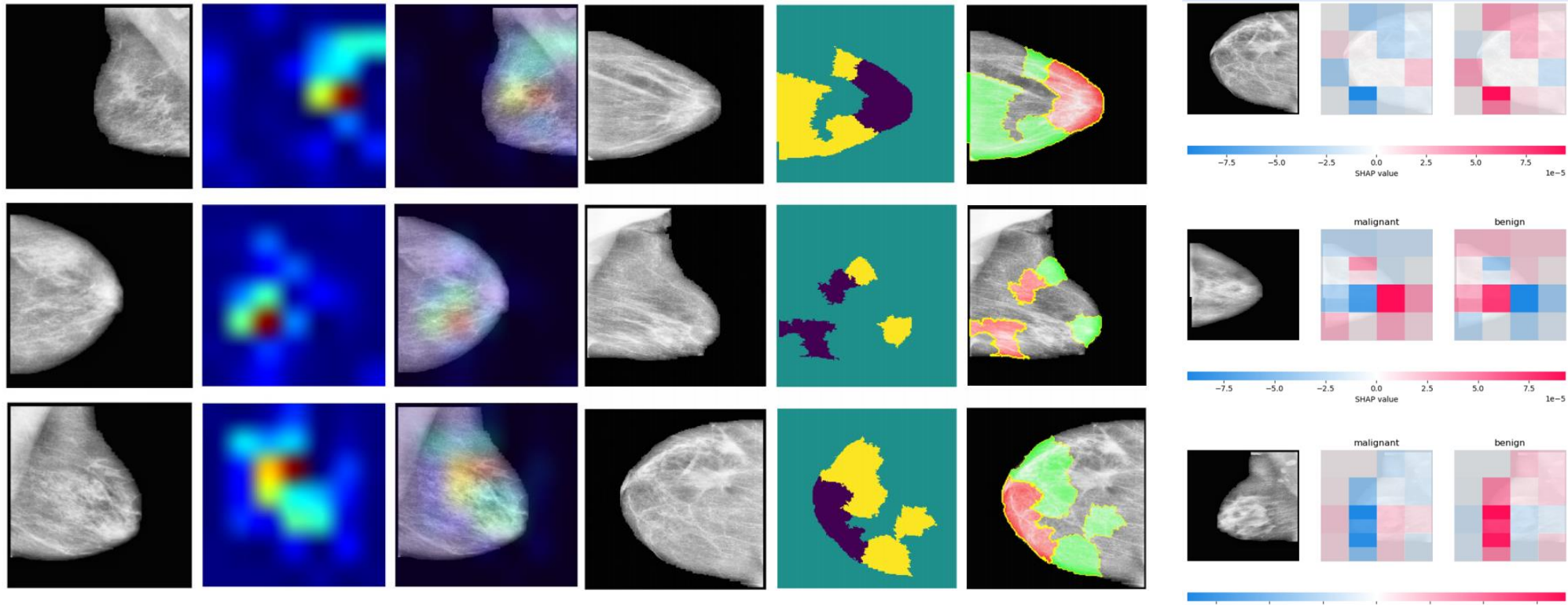


Network	F1 score	ACC	AUC	SEN	SPE
ResNet18	0.9485	0.9107	0.9565	0.9136	0.8850
ResNet50	0.9467	0.9080	0.9627	0.9083	0.9050
EfficientNet-B0	0.9402	0.8970	0.9486	0.8989	0.8800
DenseNet121	0.8955	0.8283	0.9367	0.8181	0.9200
Proposed Ensemble	0.9523	0.9173	0.9634	0.9183	0.9075

- Mammographic density classification (A, B, C, D) using a combined model (ResNet18, ResNet50, EfficientNet-B0, DenseNet121)
- Dataset: VinDr-Mammo
- Accuracy: AUC: 0.963, F1-score: 0.952 (Soft voting)

Fig 5. Attention-Enhanced Deep Learning Ensemble for Breast Density Classification in Mammography

2. Related works



- Model: VGG16, InceptionV3, Res-Net18
- Dataset: CIBS-DDSM
- Accuracy: 76%

Fig 6. Enhancing Breast Cancer Diagnosis in Mammography: Evaluation and Integration of Convolutional Neural Networks and Explainable AI (Ahmed et al. 2024)

2. Related works

- **Hard/soft voting:**

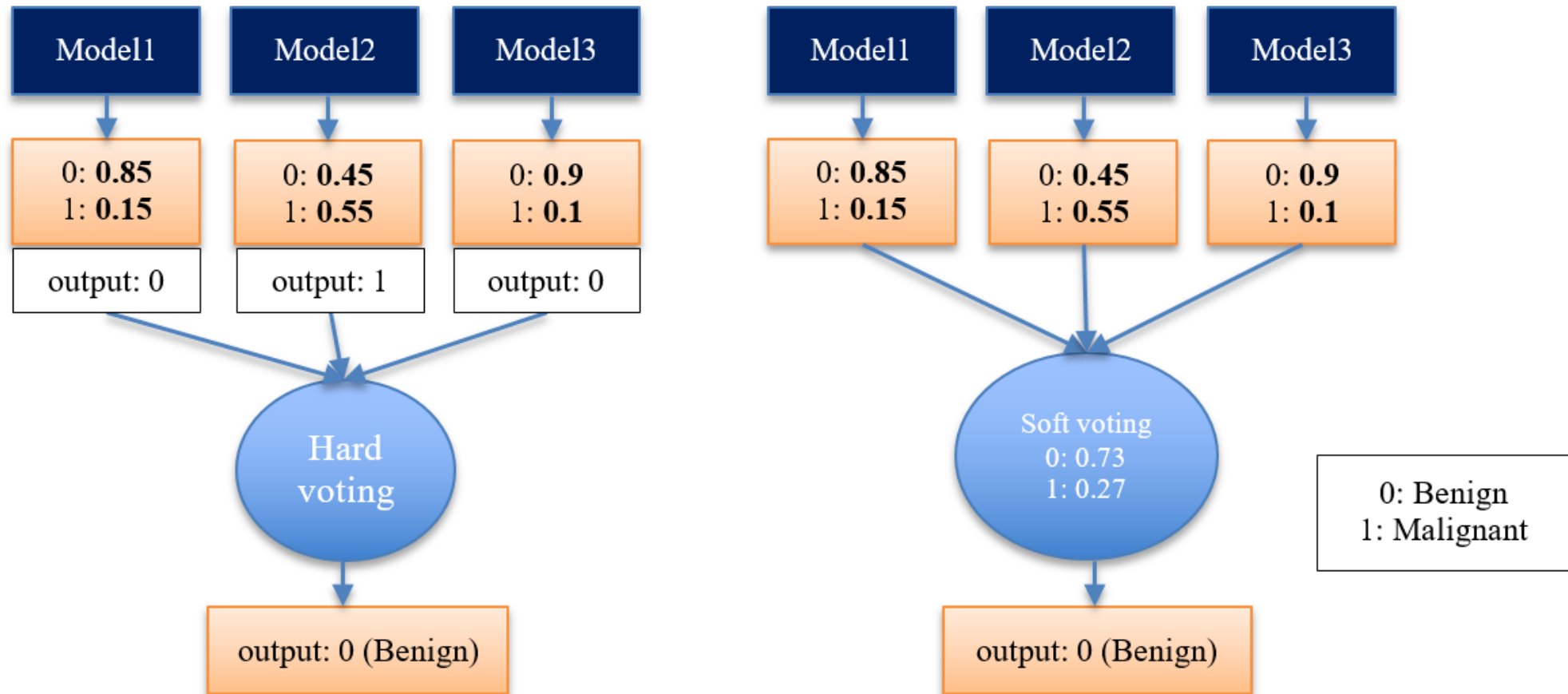


Fig 7. Illustration of Hard voting and Soft voting

2. Related works

- **Evaluation metrics:**

$$Acc = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i$$

$$Recall = \frac{TP}{TP + FN}$$

$$(1) \quad \hat{t}_{weight.hard} = \arg \max_c \left(\sum_{i=1}^k w_i \cdot 1_{\{m_i(x)=c\}} \right) \quad (2)$$

$$(3) \quad \hat{t}_{weight.soft} = \arg \max_c \left(\sum_{i=1}^n w_i \cdot p_i(c|x) \right) \quad (4)$$

$$(5) \quad F1 - score = 2x \frac{Precision * Recall}{Precision + Recall} \quad (6)$$

$$(7)$$

3. Proposed Method

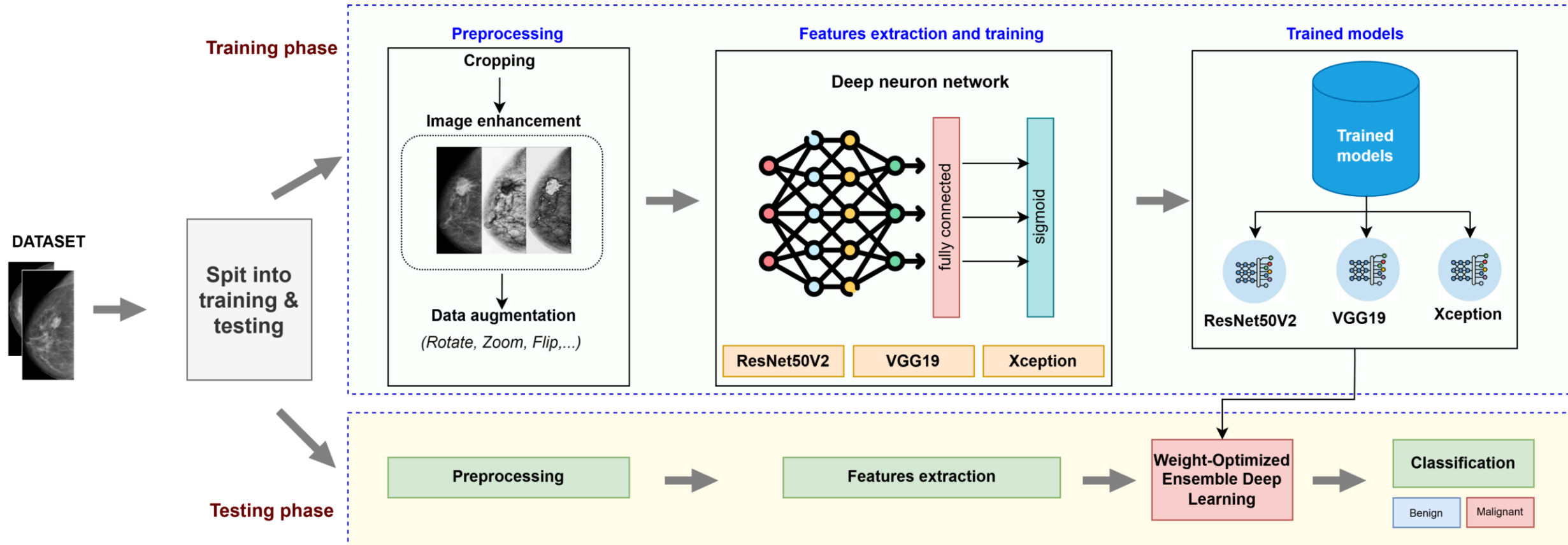


Fig 8. A general model is proposed for breast cancer classification

3. Proposed Method

- **Data preprocessing:** the input data is preprocessed using a number of techniques such as cropping, image enhancement, and data augmentation (flip, rotate, zoom, etc.).
- **Training phase:** Each model is trained independently on the same dataset and unified parameters. We use the early stopping technique during training to limit the model from overfitting. The best training results of each model are saved for testing.
- **Testing phase:** The test image is processed similarly through data preprocessing steps, feature extraction, then hard voting and soft voting techniques are applied, combined with weights optimized by Optuna technique on 3 models, to give the final result as benign or malignant.

3. Proposed Method

Algorithm 1 Weight-Optimized Ensemble Deep Learning Algorithm

- 1: **Input:** A set of trained models $\{t_1, t_2, \dots, t_i\}$; test dataset X_{test} ; ground-truth labels y_{test} ; voting scheme $\text{mode} \in \{\text{soft}, \text{hard}\}$
 - 2: Retrieve the prediction scores from each model m_i
 - 3: Create an case study with the objective of maximizing classification accuracy
 - 4: **for** each trial **do**
 - 5: Randomly assign weights $z_1, z_2, \dots, z_m \in [0, 1]$
 - 6: Normalize weights: $z_i \leftarrow \frac{z_i}{\sum_{j=1}^m w_j}$
 - 7: **if** $\text{mode} = \text{hard}$ **then**
 - 8: Aggregate predictions using the hard voting rule (Eq. 2)
 - 9: Produce the predicted labels $\hat{l}$
 - 10: **else if** $\text{mode} = \text{soft}$ **then**
 - 11: Aggregate predictions using the soft weighted rule (Eq. 4)
 - 12: Produce the predicted labels $\hat{l}$
 - 13: **end if**
 - 14: Evaluate $\text{Accuracy} = \text{acc}(y_{\text{test}}, \hat{l})$
 - 15: Report Acc back to the case study
 - 16: **end for**
 - 17: Select the best-performing weight set z_i^* from the highest scoring trial
 - 18: **return** predictions $\hat{l}$ obtained with z_i^*
 - 19: **Output:** Final prediction $\hat{l}$, ptimal weights z_i^*
-

4. Experimental results

- **Environment:** The system runs in a Python environment on Kaggle, using an Intel Xeon 2.00 GHz processor with 30GB RAM. It also utilizes a graphics processor with a 16GB RAM configuration and the Nvidia Tesla P100 GPU. Tensorflow v1.5 is the library used for network model training.
- **Dataset:** The dataset is compiled from the VinDr dataset, supplemented with data from several hospitals. The number of images we use in the paper includes 6930 images (5862 images were used for training and 1068 images for testing). The dataset is divided in the ratio of 80:20.

Table 1: Proposed scenarios and training parameters

Sce.	Model	Epochs	Batch size	Learning rate	Optimizer
1	ResNet50V2	100	32	0.0001	Adam
2	VGG19	100	32	0.0001	Adam
3	Xception	100	32	0.0001	Adam
4	Ensemble Deep Learning (Hard voting)				
5	Ensemble Deep Learning (Soft voting)				
6	Weight-Optimized Ensemble (Hard Voting)				
7	Weight-Optimized Ensemble (Soft voting)				

4. Experimental results

- Training results

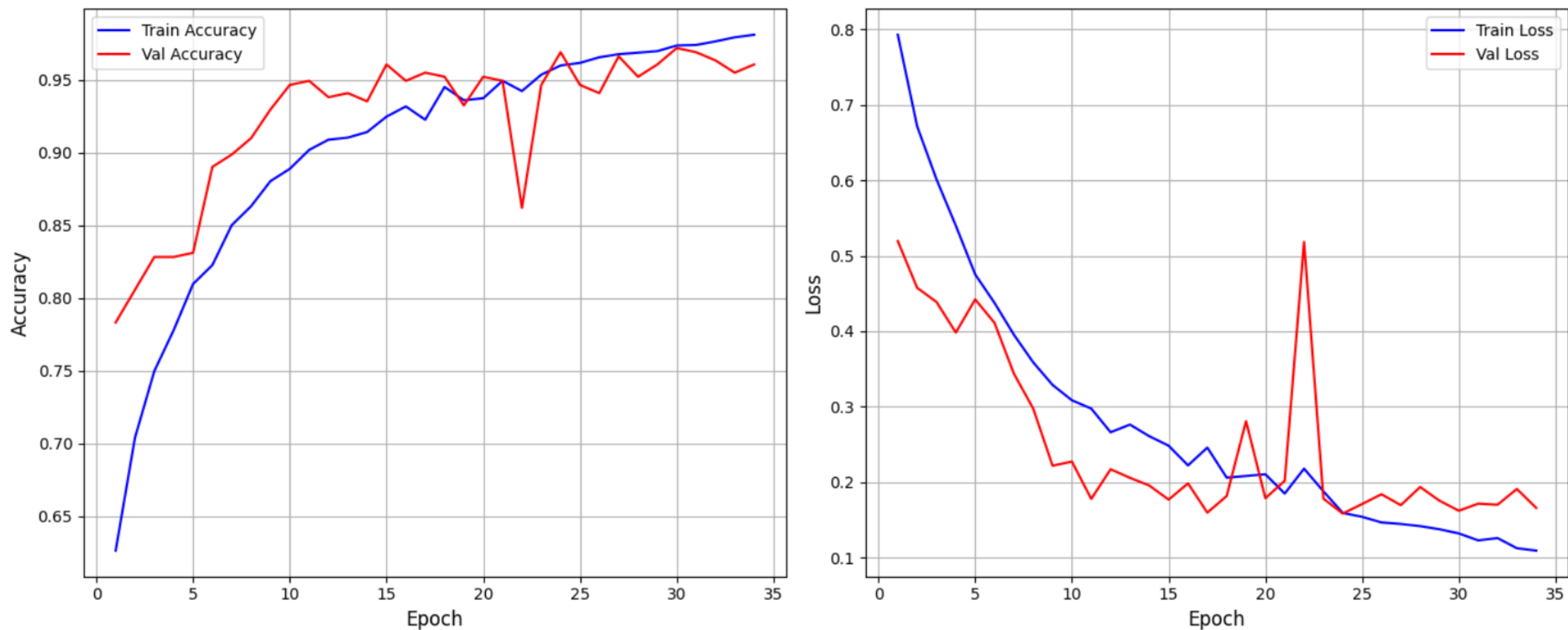


Fig 9a. The chart compares the accuracy and loss of Sce.1.

4. Experimental results

- Training results

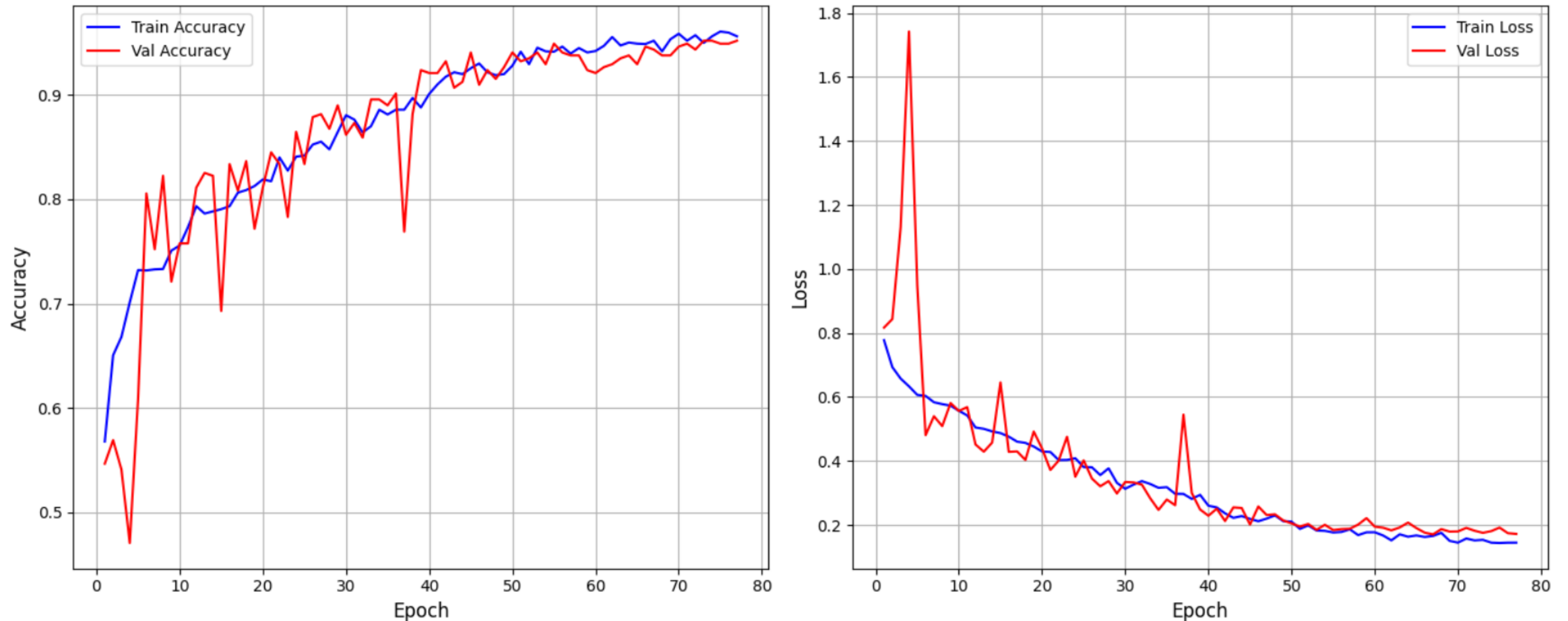


Fig 9b. The chart compares the accuracy and loss of Sce.2.

4. Experimental results

- Training results

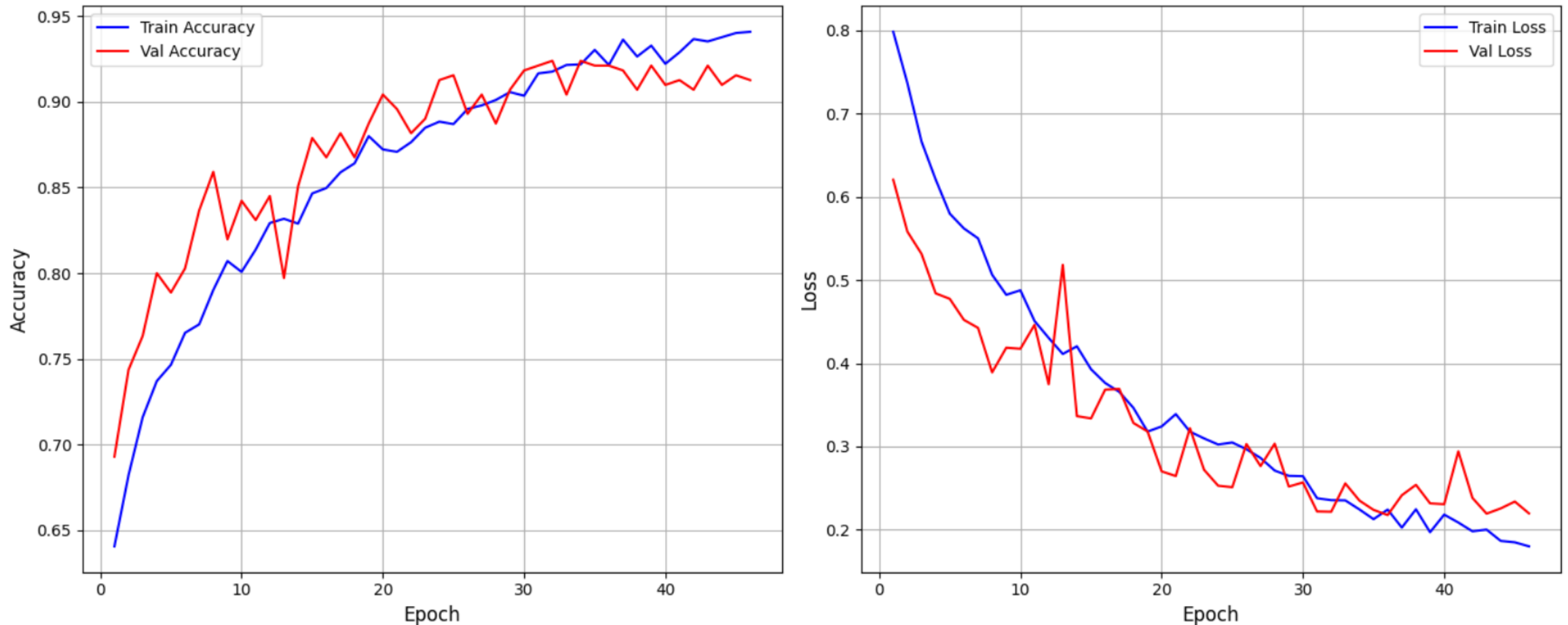


Fig 9c. The chart compares the accuracy and loss of Sce.3.

4. Experimental results

- Training results

Scenario	Accuracy (%)	Loss	Time (minutes)
Scenario 1	95.1	0.190	156.81
Scenario 2	92.3	0.232	410.25
Scenario 3	91.7	0.246	214.56

4. Experimental results

Correlation of parameters (Hard Voting)

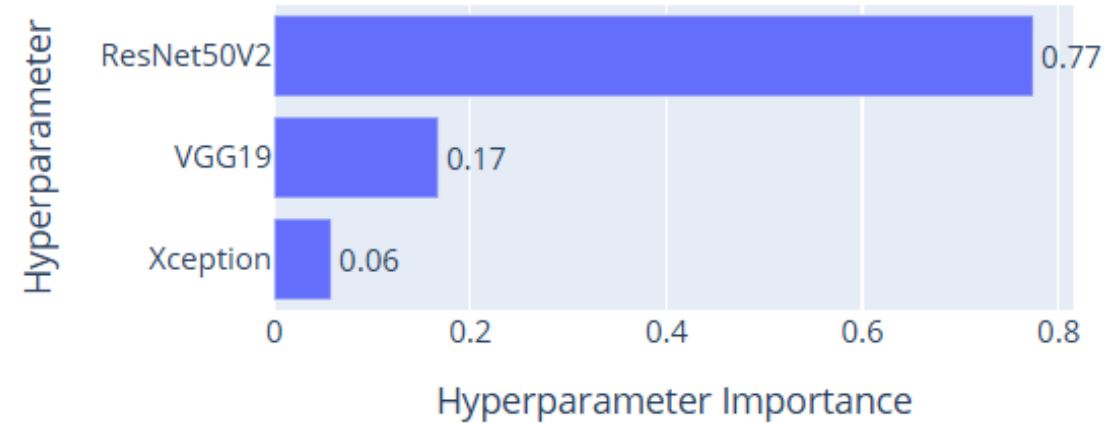
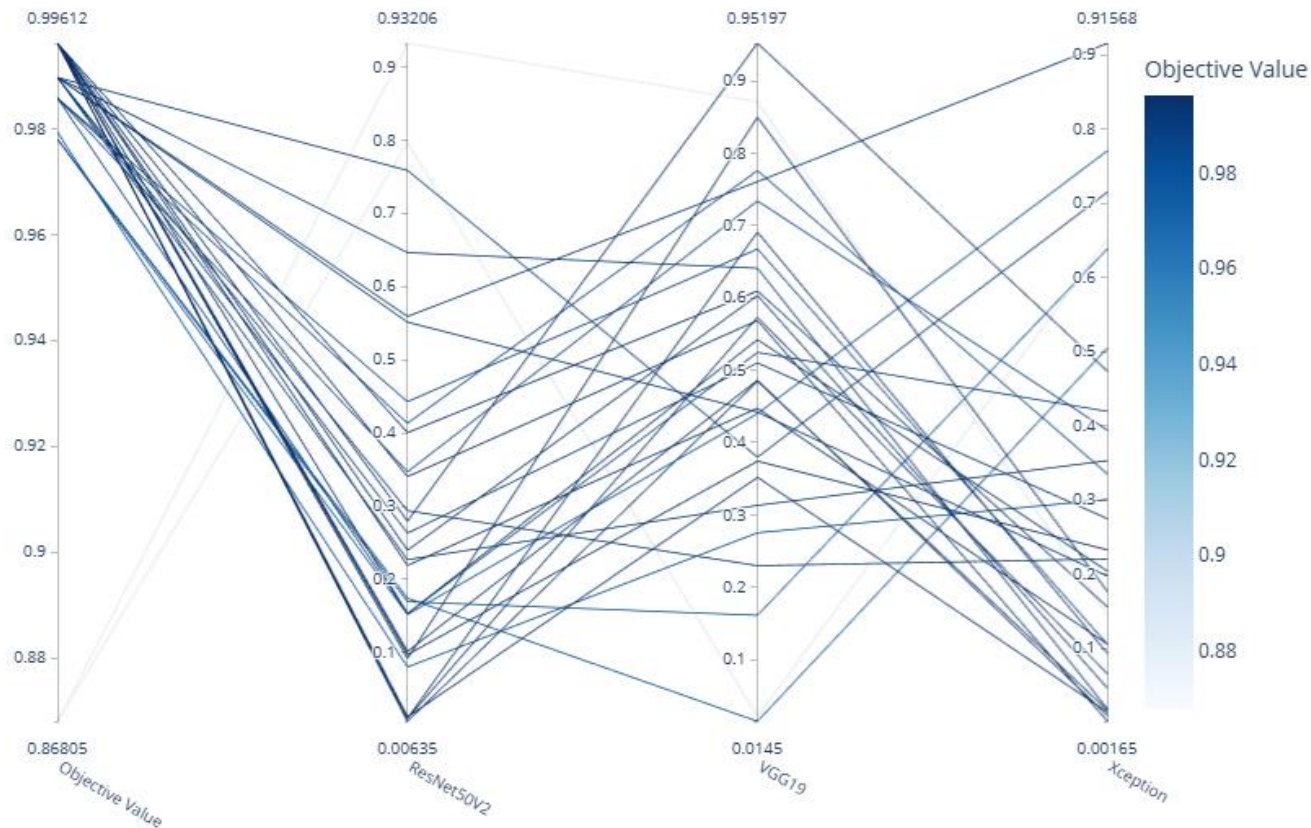


Fig 10a. Correlation and importance of parameters in Sce. 6.

4. Experimental results

Correlation of parameters (Soft Voting)

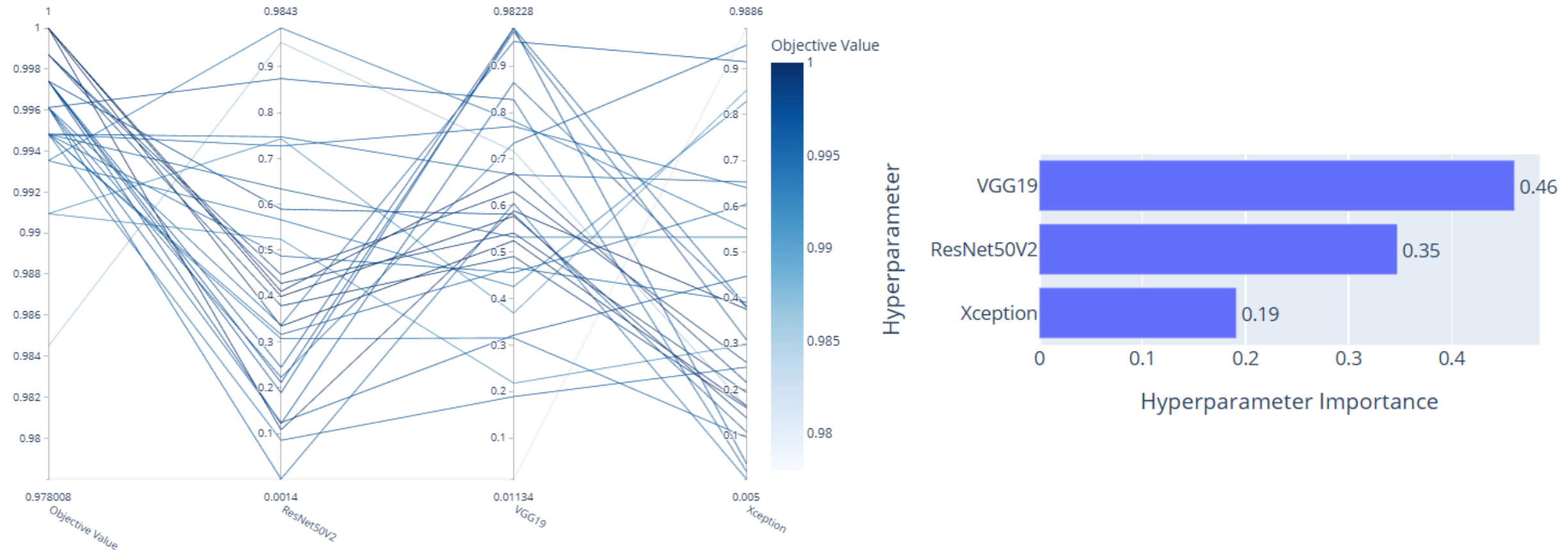


Fig 10b. Correlation and importance of parameters in Sce. 7.

4. Experimental results

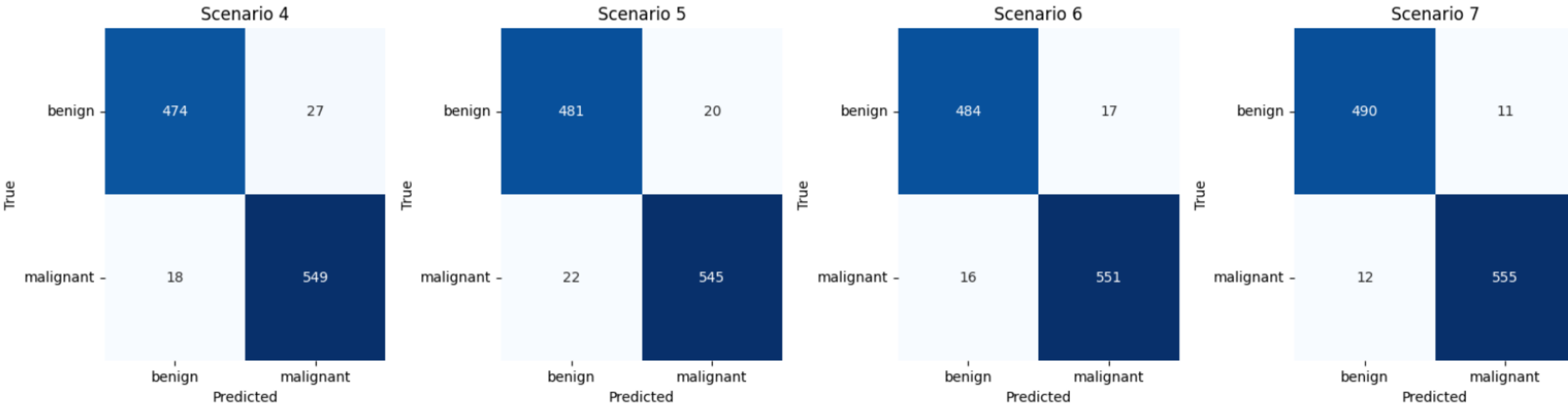


Fig 11. Confusion matrix of Sce. 4, 5, 6, 7

4. Experimental results

Tab 2. Evaluation results on test dataset

Scenario	Classify	Precision(%)	Recall(%)	F1-score(%)	mAP(%)
Scenario 1	B	95.81	94.67	95.24	95.51
	M	95.24	96.26	95.74	
Scenario 2	B	91.02	93.25	92.12	92.70
	M	94.18	92.23	93.19	
Scenario 3	B	88.82	92.32	90.54	91.29
	M	93.47	90.44	91.93	
Scenario 4	B	94.61	96.34	95.47	95.79
	M	96.83	95.31	96.06	
Scenario 5	B	96.01	95.63	95.82	96.07
	M	96.12	96.46	96.29	
Scenario 6	B	96.61	96.8	96.7	96.91
	M	97.18	97.01	97.09	
Scenario 7	B	97.8	97.61	97.71	97.85
	M	97.88	98.06	97.97	

4. Experimental results

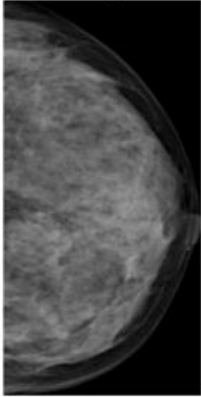
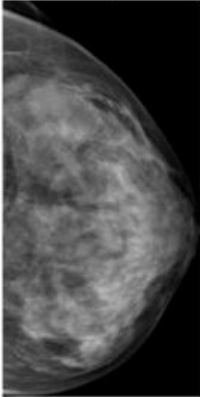
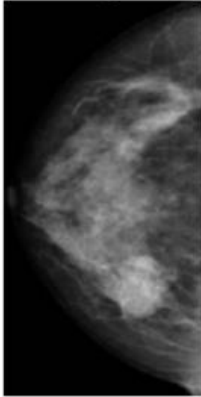
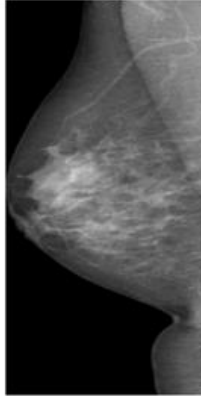
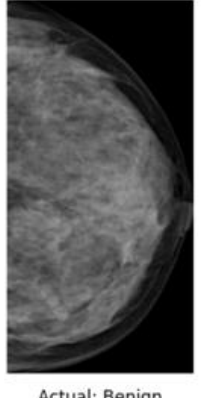
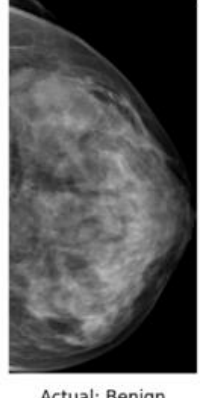
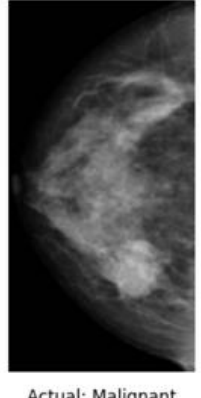
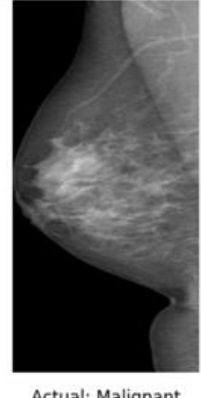
Method	Output			
	Img1	Img2	Img3	Img4
Scenario 6				
	Actual: Benign Predict: Benign Voting: 0/3 Malignant 3/3 Benign	Actual: Benign Predict: Benign Voting: 0/3 Malignant 3/3 Benign	Actual: Malignant Predict: Malignant Voting: 3/3 Malignant 0/3 Benign	Actual: Malignant Predict: Malignant Voting: 3/3 Malignant 0/3 Benign
Scenario 7				
	Actual: Benign Predict: Benign p_soft(c=0): 1.00 p_soft(c=1): 0.00	Actual: Benign Predict: Benign p_soft(c=0): 1.00 p_soft(c=1): 0.00	Actual: Malignant Predict: Malignant p_soft(c=0): 0.00 p_soft(c=1): 1.00	Actual: Malignant Predict: Malignant p_soft(c=0): 0.00 p_soft(c=1): 1.00

Fig 12. Prediction results on the test dataset of Sce. 6 and 7.

5. Conclusion

- We propose a comprehensive method for detecting and classifying breast cancer using deep learning ensemble models.
- We propose optimizing the contribution weight of each component model to improve classification performance.
- The results show that the combined model achieved 97.85% accuracy on the test dataset, while significantly reducing the misclassification phenomenon in both groups: benign and malignant lesions.

Future works:

- Expanding the research by evaluating more advanced deep learning models to determine the most optimal combination.
- Focus on multi-class classification and integrating more specialized mammographic features, such as tissue density and the presence of microcalcifications, to further improve the accuracy and coverage when applied to more diverse and complex datasets.

References

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